

Sentient.io: Democratizing AI for global developers with BigQuery and GKE

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ABOUT SENTIENT.IO

Sentient.io create ML solutions, save time in big data and achieves 100% boost in product release speed and boosts client success with Google Cloud



About Sentient.io

Sentient.io is a startup on a mission to make artificial intelligence (AI) more accessible by enabling companies to deploy AI in their business with no prior machine learning (ML) experience. The platform develops AI algorithms that developers launch out of the box for easy, intuitive ML integration. From May 2017 till today, Sentient.io has raised a total of USD\$5million.

**Google
Cloud
results**



100%

improvement

Industries: Technology

- Improves productivity by yielding 90% time savings in data queries with BigQuery
- Doubles product release speed by automating ML modeling with BigQuery
- Powers NLP solution for Singapore client that yielded six-fold savings in manpower hours
- Enables fault detection tool for a Japanese manufacturer with 98% accuracy rates

product cycle

Location: Singapore

Cloud Pub/Sub

Google Cloud (<https://cloud.google.com/>)

Container

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage

Dataflow (<https://cloud.google.com/dataflow/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

TENSORFLOW (<https://www.tensorflow.org/>)

Container Registry

(<https://cloud.google.com/container-registry/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Let us take on your challenge. We're here to help.

Cloud Speech to Text

(<https://cloud.google.com/speech/>)

Cloud Text to Speech

(<https://cloud.google.com/text-to-speech/>)

TensorFlow (<https://www.tensorflow.org/>)

As a Star Trek fan in the 1970s, Christopher Yeo grew up in Singapore with dreams of empowering intelligent life through technology. Today he's building artificial intelligence (AI) solutions that might not seem out of place on Starship Enterprise.

Three years ago, Christopher launched [Sentient.io](https://sentient.io/en/) (<https://sentient.io/en/>), a platform that aims to make AI accessible by giving developers with no prior machine learning experience pre-trained AI algorithms to deploy in their apps. The goal is to contribute to humankind by bringing the rarefied realms of voice AI, computer vision, big data analytics, and natural language processing down to earth.

"Our goal is to
use AI to help
people with big
ideas develop
their own
groundbreaking
solutions. And
the relationship
with Google
Cloud goes
beyond enabling
our algorithms

with features
such as Google
Kubernetes
Engine and
BigQuery. We're
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by creative
collaboration
with Google
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ML solutions to
build a better
world."

*—Christopher Yeo,
founder and CEO,
Sentient.io*

Christopher and his team had decades of advanced AI acumen under their belt. To take flight, however, they needed access to powerful machine learning and computing tools. They turned to Google Cloud (<https://cloud.google.com/>) to enable a long-term ambition to reach one billion lives within five years. Sentient.io's impact ranges from voice recognition in Singapore's vibrant linguistic melting pot to defect detection in Japan's ultra high-precision manufacturing sector.

"Our goal is to use AI to help people with big ideas develop their own groundbreaking solutions," says Christopher. "And the relationship with Google Cloud goes beyond enabling our algorithms with features such as [Google Kubernetes Engine](#) (/kubernetes-engine) and [BigQuery](#) (/bigquery). We're equally inspired by creative collaboration with Google Cloud to drive ML solutions to build a better world."

Finding the right partner to bring AI to the world's fingertips

Christopher's journey with Google Cloud began last year when he attended a [Google Next](#) (<https://cloud.withgoogle.com/next/sf/>) conference in Singapore with his team. In the two years since launching, Sentient.io had been with a different cloud provider and was seeking ways to take its AI solutions to a higher level.

At Google Next, Christopher says he found more than superior ML capability. What impressed him most was a kindred spirit in developing AI in unexpected, life-enhancing, and collaborative ways. "When we met Google Cloud," says Christopher, "we found a shared humanistic AI outlook in our DNA. That's fundamentally what sparked our journey together."

The eye-opening encounter led Sentient.io to begin moving its services little by little to Google Cloud. It started with BigQuery then adopting Google Kubernetes Engine (GKE) for computing

infrastructure. What began as a piecemeal move snowballed into rapid full-fledged migration.

"We started moving stuff, and since it worked great, we'd say, 'Okay, let's move that as well.' Then 'why not move this too?' And before we knew it, we'd moved more than 80% of our platform to Google Cloud," Christopher says.

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AI performance and productivity gains with BigQuery

The swift migration would not have happened without concrete Google Cloud performance metrics. Sentient.io found that BigQuery enabled it to obtain 90% time savings in data queries for projects such as the analysis of telco location data for retail analytics.

That was due to intuitive interface and automated ML fine-tuning, even dealing with petabytes of data. "In the old days, we had to manually manage big sets of data. So my team members spent a lot of time fiddling with data structures and performance issues," recalls Christopher. "Discovering BigQuery was a real game changer. We said wow, this is so simple. We just throw in the data and BigQuery does the rest."

AutoML yielded similar performance gains and time savings. The enhancements freed up Christopher's

team to focus on creative tasks, rather than mundane and repetitive model training. "In the past, one of our AI engineers would create an ML model from scratch, then start training and tuning, over and over. If they couldn't figure out how to get the best performance, they'd start from scratch and create a new model," says Christopher. "Now, using AutoML, we simply tell the program what to do. These are my computer vision images, for example, and here's the performance I want. AutoML then goes away to discover neural network topologies that work best for that data."

How much time has AutoML saved Sentient.io's team? Christopher estimates AutoML takes over roughly 90% of routine training and tuning tasks, enabling data scientists to fill that time with creative endeavor. "We work on monthly product cycles, releasing a feature set each month," says Christopher. "In the past, it took us two cycles to release 30 feature sets. Now we can release 30 feature sets in one month, cutting that time in half."



Simplifying speech-to-text for Singapore's linguistic melting pot

Sentient.io's imaginative, no-limits approach to AI development melds with Google Cloud solutions to achieve remarkable outcomes. For example, Sentient.io worked with Google Cloud to create English subtitling and transcription services for Singapore's national broadcaster [Mediacorp](https://www.mediacorp.sg/en/) (<https://www.mediacorp.sg/en/>).

In Singapore's vibrant ethnic mix that includes Chinese, Malay, Indian, Peranakan, European, and Nepali, among others, English has become the commonly spoken language. Individuals from each ethnic group speak English with different accents, and pepper their speech with words, even grammatical structures, from their native tongue. This makes it a challenge to create reliable voice-to-text.

"Since Singapore has so many ethnic groups, the computer system just goes crazy if you run standard speech recognition systems here," Christopher explains. "A single English sentence can have four different words that come from different languages."

To solve this issue, Sentient.io worked with IMDA and used the National Speech Corpus which collected local voice assets not only from each ethnic group, but subsets within each group,

according to factors such as gender, age, and education. That adds up literally to 3,000 hours of speech samples across Singapore's linguistic range. It then deployed these assets in an effective Speech algorithm. Google Cloud provided the framework for Sentient.io's innovation to come to life. The result? MediaCorp increased its transcription speed six-fold using Sentient.io's speech-to-text solution.

In another AI achievement, Sentient.io developed several deep learning models and also used Google AutoML to improve defect detection and achieved 0% false acceptance and 98.34% false rejection rate for a specialty steel manufacturer in Japan. That may not sound like much, but considering the client [Daido Steel](https://www.daido.co.jp/en/) (<https://www.daido.co.jp/en/>) makes precision parts, a 5% improvement can mean the difference between life and death.

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Partnering with Google Cloud beta features to build AI's future

What lies ahead in Sentient.io's ambition to reach one billion lives? First, it plans to vastly expand its library of out-of-box AI algorithms. Next it's going beyond developer solutions to craft AI for end users, in particular people with learning and physical disabilities. For these aims, Sentient.io is eager to experiment with the beta features of Google Brain (<https://research.google/teams/brain/>). And Christopher is impressed by Google Cloud's own eagerness to share experimental solutions in a collaborative way.

"We're constantly in touch with our Google Cloud partners in Asia, and they've hooked us up with researchers at Google Mountain View headquarters," says Christopher. "This information sharing, this getting to know each other, it's hugely important to us."

Christopher sums up Sentient.io's relationship with Google Cloud as a human touch in machine intelligence. "When you get a bunch of smart people together from both teams simply working on cool ideas, really amazing, creative things happen," he says. "And it's so friendly. We all enjoy building new solutions while going out together for drinks."